Online PES Decoder Learning in Neuromorphic Residual Control of UAV Quaternion Tracking

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***Abstract*— Online neuromorphic residual control can augment a classical attitude controller without changing its feedback gains or actuator interface. Six continuous coupled membrane-gate (CMG) units and eighteen decoder weights adapted by prescribed error sensitivity (PES) add a bounded correction to the saturated quaternion proportional-derivative (PD) torque request before rotor-force allocation. Across ten matched 45-s MuJoCo cases on the Skydio X2 and a Crazyflie 2 derived model, mean quaternion-angle error falls by 17.3–60.6% (53–57% on the Full3 and Hard45 multi-axis missions). The same units with a fixed decoder give at most 10.3%; their internal signals attribute the gain to decoder adaptation rather than to the neuron dynamics alone. Against LIF, ALIF, Izhikevich, and LIFRate bases, CMG–PES attains better tracking at substantially lower realized residual effort. The reductions persist under synthetic measurement noise, and learned corrections are reused across repeated disturbances. PD remains a nominal, interpretable reference; the residual adds bounded, revocable adaptation.**

I. Introduction

Learning-based augmentation can retain classical feedback while adapting a correction that is difficult to specify *a priori*. We study a bounded torque residual for quaternion attitude tracking of an unmanned aerial vehicle (UAV) with individual rotor-force saturation. The question is whether a small stateful basis improves a fixed PD controller, and whether any improvement comes from its neuron dynamics, decoder learning, or correction magnitude. Bounded authority makes the learning layer switchable; it does not by itself ensure stability or safety.

Prior work includes residual robot learning [1], geometric $\mathcal{L}_1$ quadrotor augmentation with performance guarantees [2], and proportional–integral–derivative (PID) attitude control with a proximal policy optimization (PPO) residual [3]. Neuromorphic control offers a complementary implementation: Nengo supports the Neural Engineering Framework (NEF) and online decoder learning [4], most often through PES, an established local, online error-driven update [5]. PES-based adaptive control has been demonstrated on manipulators [6], [7] and in a Nengo spiking PID controller for a free-flying UAV [8]. We focus instead on a continuous membrane-gate basis for a bounded quaternion-torque residual, with a fixed classical controller and adaptation confined to its decoder.

We augment a quaternion PD baseline with coupled conductance-based membrane-gate (CMG) units and PES decoder learning. The paper's contributions are:

1) a code-aligned CMG–PES architecture with explicit paired-current encoding, state-update, and readout equations and a fully specified actuator interface (Fig. 1);
2) proofs of internal-state, command-deviation, and weight boundedness, exact PD recovery at zero authority, and an allocation-aware local stability condition;
3) a controlled ablation separating conductance modulation, decoder learning, and output authority, which traces the gain to decoder adaptation through the membrane, gate, activity, residual, and decoder signals of matched fixed- and adaptive-decoder runs; and
4) robustness evidence under measurement noise and a hold-equilibrium study in which the learned correction is reused across repeated disturbances.

The CMG unit couples a leaky membrane to a gating state that modulates its conductance, linking LIF-like integration with Hodgkin–Huxley inspired gating; the controller uses its continuous, non-spiking trajectory (related regimes can spike) as a compact set of stateful features. The implementation is digital and claims no neuromorphic-hardware energy advantage.

II. Attitude Baseline and Actuator Interface

The objective is to reduce shortest-path attitude error for a prescribed $q_d(t)$ while respecting rotor-force limits. Baseline and augmented controllers share the state measurements, height loop, and allocator; Section III modifies only the requested attitude torque.

A. Quaternion Convention and Error

Let $Q \in \mathbb{S}^3$ denote the scalar-first Hamilton quaternion representing attitude (body to inertial), and let $\boldsymbol{\omega}$ be the body angular velocity. For a rigid body with inertia $\boldsymbol{J} \succ 0$ about the center of mass, expressed in body axes (cf. Fig. 2), the free-flight rotational dynamics are [9]

$$\dot{Q} = \tfrac{1}{2} Q \otimes [0, \boldsymbol{\omega}^{\mathsf{T}}]^{\mathsf{T}}, \tag{1}$$

$$\boldsymbol{J}\dot{\boldsymbol{\omega}} = \boldsymbol{\tau}_{\mathrm{act}} + \boldsymbol{\tau}_{\mathrm{p}} + \boldsymbol{\tau}_{\mathrm{d}} - \boldsymbol{\omega} \times (\boldsymbol{J}\boldsymbol{\omega}). \tag{2}$$

Here $\boldsymbol{\tau}_{\mathrm{p}}$ is modeled passive torque and $\boldsymbol{\tau}_{\mathrm{d}}$ any applied disturbance, both zero in the nominal model; contact, if present, adds an external wrench.

We work with the corresponding passive quaternion $q = Q^{-1}$ [10], which satisfies $\dot{q} = -\frac{1}{2}[0, \boldsymbol{\omega}^{\mathsf{T}}]^{\mathsf{T}} \otimes q$; the desired

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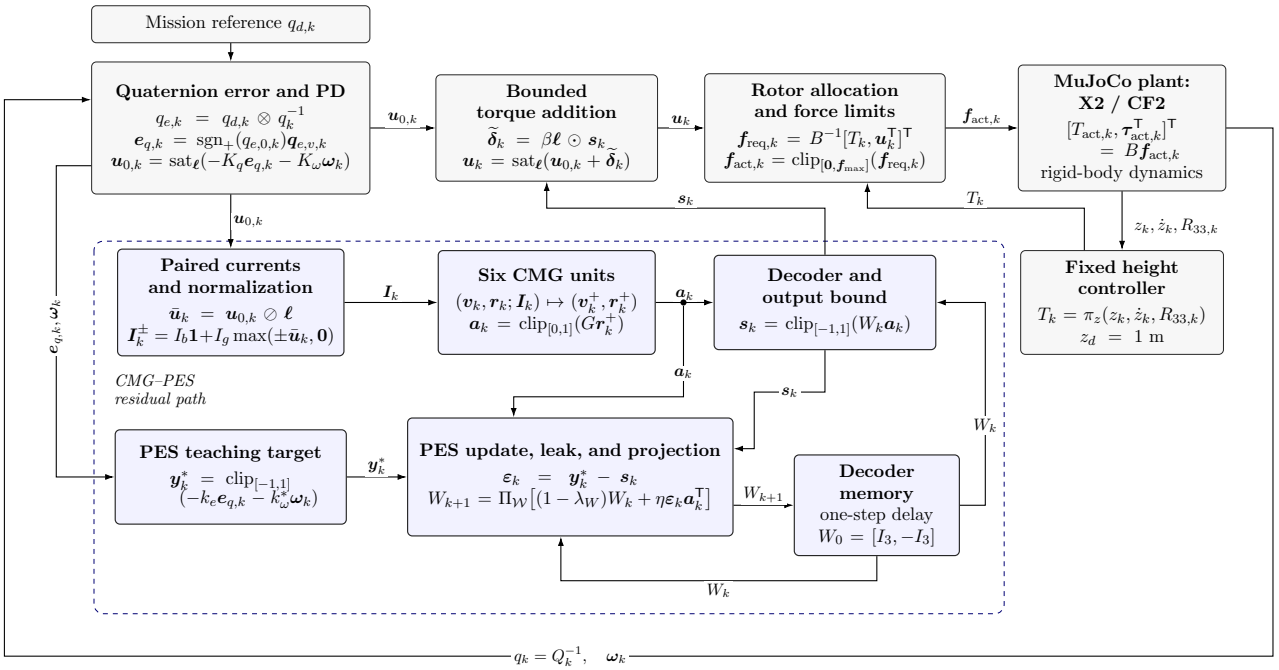


Fig. 1. Online PES decoder learning CMG-PES Neuromorphic Residual Control architecture



Fig. 2. CF2 (left) and X2 (centre) hardware and MuJoCo models, and both geometries at relative scale (right). Body-frame axis triads rotate with the vehicle relative to the inertial frame (cf. Q: body $\to$ inertial, Sec. II).

attitude q_d uses the same convention. Define the attitude error

$$q_e = q_d \otimes q^{-1} = [q_{e,0}, \boldsymbol{q}_{e,v}^\mathsf{T}]^\mathsf{T}, \tag{3}$$

$$\boldsymbol{e}_q = \mathrm{sgn}_+(q_{e,0}) \boldsymbol{q}_{e,v}, \qquad \mathrm{sgn}_+(0) = 1, \tag{4}$$

$$\theta_q = 2 \operatorname{atan2}(\|\boldsymbol{q}_{e,v}\|_2, |q_{e,0}|) \in [0, \pi]. \tag{5}$$

The reported θ_q is the shortest rotation angle, invariant under $q_e \mapsto -q_e$; sign selection introduces a discontinuity at $q_{e,0} = 0$ and does not establish robust hybrid global stability [11].

For a desired-frame rate $\boldsymbol{\omega}_d$ satisfying the corresponding passive kinematics, differentiation gives

$$\dot{q}_e = \tfrac{1}{2} q_e \otimes [0, \boldsymbol{\omega}_e^\mathsf{T}]^\mathsf{T}, \tag{6}$$

$$\boldsymbol{\omega}_e = \boldsymbol{\omega} - R(q_e)^\mathsf{T} \boldsymbol{\omega}_d, \qquad \dot{q}_{e,0} = -\tfrac{1}{2} \boldsymbol{q}_{e,v}^\mathsf{T} \boldsymbol{\omega}_e. \tag{7}$$

These kinematic identities imply no convergence; reference Euler triples parameterize q_d in this convention.

B. Physical Inertia and Implemented PD Gains

When principal moments $\boldsymbol{j}_\mathrm{p}$ of $\boldsymbol{J}$ are specified in a rotated principal-axis frame, the body-frame tensor is $\boldsymbol{J} = R_{BI} \operatorname{diag}(\boldsymbol{j}_\mathrm{p}) R_{BI}^\mathsf{T}$ for the rotation R_{BI} from principal to body axes. The implementation sets axiswise gains from the principal moments in their reported ordering, $\boldsymbol{j}_\mathrm{tune}$, not necessarily $\operatorname{diag}(\boldsymbol{J})$ (Table I).

The fixed quaternion PD baseline is

$$\boldsymbol{k}_q = 2 \boldsymbol{j}_\mathrm{tune} \odot \boldsymbol{\omega}_n^2, \qquad \boldsymbol{k}_\omega = 2 \zeta \boldsymbol{j}_\mathrm{tune} \odot \boldsymbol{\omega}_n, \tag{8}$$

$$\boldsymbol{u}_0 = \mathrm{sat}_\ell(-\boldsymbol{k}_q \odot \boldsymbol{e}_q - \boldsymbol{k}_\omega \odot \boldsymbol{\omega}), \tag{9}$$

where $\odot$ denotes componentwise multiplication, $\boldsymbol{\omega}_n = (4, 4, 3.5)^\mathsf{T}$ rad/s, $\zeta = 0.95$, and sat_ℓ clips component i to $[-\ell_i, \ell_i]$. The componentwise torque limits $\boldsymbol{\ell} \in \mathbb{R}^3_{>0}$ define the requested-torque box $\mathcal{U} = \prod_{i=1}^3 [-\ell_i, \ell_i]$, with $\ell_i = \max(j_{\mathrm{tune},i} \nu_i, 10^{-6}\,\mathrm{N\,m})$ and acceleration scale $\boldsymbol{\nu} = (12, 12, 10)^\mathsf{T}$ rad/s^2; these controller-side bounds are not the rotor-feasible torque set. $\mathrm{clip}_{[a,b]}$ also acts componentwise, and $\oslash$ is componentwise division. The law is quaternion feedback [12]; $\boldsymbol{\omega}_n$ is a tuning parameter, not a set of coupled closed-loop natural frequencies. Desired-rate feedforward and gyroscopic cancellation are omitted. The gains are vehicle-specific through $\boldsymbol{j}_\mathrm{tune}$ (with $\boldsymbol{\omega}_n, \zeta$ shared), identical for PD and its CMG augmentation, and used unchanged, not as a scalar-gain surrogate, in Section III-D.

C. Requested and Realized Wrench

For rotor positions $\boldsymbol{p}_i$ in body coordinates, parallel $+z$ thrust axes, and signed reaction coefficients κ_i, define

$$B = \begin{bmatrix} 1 & 1 & 1 & 1 \\ p_{1,y} & p_{2,y} & p_{3,y} & p_{4,y} \\ -p_{1,x} & -p_{2,x} & -p_{3,x} & -p_{4,x} \\ \kappa_1 & \kappa_2 & \kappa_3 & \kappa_4 \end{bmatrix}. \tag{10}$$

The signs follow $\boldsymbol{p}_i \times [0, 0, f_i]^\mathsf{T}$ and the local reaction torque $\kappa_i f_i$ (κ_i in meters); B maps rotor thrusts to total thrust and body torque. For collective thrust T and body-torque command $\boldsymbol{u}$, inverse allocation with clipping is

$$\begin{aligned} \boldsymbol{f}_{\mathrm{req}} &= B^{-1}[T, \boldsymbol{u}^\mathsf{T}]^\mathsf{T}, \\ \boldsymbol{f}_{\mathrm{act}} &= \mathrm{clip}_{[\boldsymbol{0}, \boldsymbol{f}_{\max}]}(\boldsymbol{f}_{\mathrm{req}}), \\ [T_{\mathrm{act}}, \boldsymbol{\tau}_{\mathrm{act}}^\mathsf{T}]^\mathsf{T} &= B\boldsymbol{f}_{\mathrm{act}}. \end{aligned} \tag{11}$$

Here $\boldsymbol{u} = \boldsymbol{u}_0$ for PD; the CMG path (Section III) modifies this request before the same allocation. Each f_i is a commanded force (N), not a rotor-speed command, and the wrench is realized exactly only if $B^{-1}[T, \boldsymbol{u}^\mathsf{T}]^\mathsf{T} \in [\boldsymbol{0}, \boldsymbol{f}_{\max}]$, a collective-dependent set that need not contain a componentwise torque box; allocation is clipped inverse, not optimization-based [13].

Collective thrust remains classical, with $z_d = 1$ m:

$$T = \mathrm{clip}_{[0, 2.5Mg]}\left(\frac{M[g + 6(z_d - z) - 4\dot{z}]}{\max(R(Q)_{33}, 0.35)}\right). \tag{12}$$

Here g is gravity; the loop reads altitude and vertical velocity, and the denominator compensates tilt. The guard 0.35 avoids near-singular compensation as $R(Q)_{33} \to 0$, limiting it to about $\arccos(0.35) \approx 69.5°$; the upper clip does not supersede individual rotor limits. The CMG path has no collective output but can affect height through attitude, tilt compensation, and allocation saturation.

III. CMG State Dynamics and Online Decoder Adaptation

A. Encoding, States, and Numerical Update

For m torque channels and $2m$ units, normalize the clipped baseline and form positive/negative current pairs:

$$\begin{aligned} \boldsymbol{I}^{\pm} &= I_b\boldsymbol{1} + I_g \max(\pm\bar{\boldsymbol{u}}, \boldsymbol{0}), \quad \bar{\boldsymbol{u}} = \boldsymbol{u}_0 \oslash \boldsymbol{\ell}, \\ \boldsymbol{I} &= [(\boldsymbol{I}^+)^\mathsf{T}, (\boldsymbol{I}^-)^\mathsf{T}]^\mathsf{T}. \end{aligned} \tag{13}$$

Each unit has a dimensionless membrane state v_j and gate r_j:

$$\dot{r}_j = [\sigma(\alpha_h(v_j - v_{\mathrm{th}} + \gamma r_j)) - r_j]/\tau_r, \tag{14}$$

$$\dot{v}_j = -(v_j - v_{\mathrm{rest}})/\tau + I_j/\tau - c_j(v_j - v_{\mathrm{reset}}), \tag{15}$$

$$c_j = (g_0 + \chi g_1 |I_j|) r_j^2, \qquad \chi \in \{0, 1\}. \tag{16}$$

The logistic gate target depends on v_j, while r_j shifts that target and modulates membrane conductance (Fig. 3), so both states participate although the decoder reads only the gate-derived activity. There is no threshold-triggered reset or spike; v_{reset} is a reversal level in a continuous conductance term, and g_0, g_1 make c_j a rate.

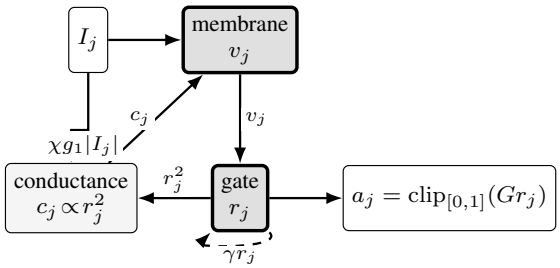


Fig. 3. Coupling topology of one CMG unit, cf. (16): the membrane sets the gate target, and the gate shifts its own threshold through γ and shunts the membrane through $c_j \propto r_j^2$ (input dependent if $\chi = 1$). The dynamics are continuous, with no spike or reset; only a_j leaves the unit.

Integration is sequential, not exponential Euler for both states: from the old states compute

$$b_{j,k} = \sigma\Big(\mathrm{clip}_{[-60,60]}[\alpha_h(v_{j,k} - v_{\mathrm{th}} + \gamma r_{j,k})]\Big), \tag{17}$$

$$r_{j,k}^+ = \mathrm{clip}_{[0,1]}[r_{j,k} + (h/\tau_r)(b_{j,k} - r_{j,k})]. \tag{18}$$

Then evaluate $c_{j,k} = (g_0 + \chi g_1 |I_{j,k}|)(r_{j,k}^+)^2$, $\lambda_{j,k} = \tau^{-1} + c_{j,k}$, and $d_{j,k} = (v_{\mathrm{rest}} + I_{j,k})/\tau + c_{j,k} v_{\mathrm{reset}}$. The frozen-coefficient membrane step is

$$v_{j,k}^+ = e^{-\lambda_{j,k}h} v_{j,k} + (1 - e^{-\lambda_{j,k}h}) d_{j,k}/\lambda_{j,k}. \tag{19}$$

It exactly solves the scalar membrane equation with frozen coefficients, without implying unconditional stability of the coupled closed loop; then $(v_{j,k+1}, r_{j,k+1}) = (v_{j,k}^+, r_{j,k}^+)$.

B. Readout, PES, and Additive Residual

The within-step activity, initial decoder, and clipped readout are

$$\begin{aligned} \boldsymbol{a}_k &= \mathrm{clip}_{[0,1]}(G\boldsymbol{r}_k^+), \qquad W_0 = [I_m, -I_m], \\ \boldsymbol{s}_k &= \mathrm{clip}_{[-1,1]}(W_k \boldsymbol{a}_k). \end{aligned} \tag{20}$$

Pair subtraction retains direction from a nonnegative basis, though state memory does not imply instantaneous sign agreement after a command reversal. With $G = 10^5$ the activity saturates once $r_j \geq 10^{-5}$, so G sets how much of r_j's range reaches the decoder. The teaching target and local update are

$$\boldsymbol{y}_k^* = \mathrm{clip}_{[-1,1]}(-k_e \boldsymbol{e}_{q,k} - k_\omega^* \boldsymbol{\omega}_k), \quad \boldsymbol{\varepsilon}_k = \boldsymbol{y}_k^* - \boldsymbol{s}_k, \tag{21}$$

$$W_{k+1} = \Pi_{\mathcal{W}}[W_k + \eta \boldsymbol{\varepsilon}_k \boldsymbol{a}_k^\mathsf{T} - \lambda_W W_k]. \tag{22}$$

Here $\mathcal{W} = [-W_{\max}, W_{\max}]^{m \times 2m}$, and η, λ_W are per step (changing h changes the learning dynamics). PES moves toward the prescribed target $\boldsymbol{y}_k^*$; it does not differentiate through the plant or minimize a closed-loop cost.

Using W_k before the update, the requested command is

$$\boldsymbol{u}_k = \mathrm{sat}_{\boldsymbol{\ell}}(\boldsymbol{u}_{0,k} + \beta \boldsymbol{\ell} \odot \boldsymbol{s}_k), \quad \boldsymbol{\delta}_k = \boldsymbol{u}_k - \boldsymbol{u}_{0,k}, \quad 0 \leq \beta \leq 1. \tag{23}$$

The canonical UAV case has $m = 3$, six CMG units, and eighteen decoder weights. Intrinsic conductance modulation (χ), synaptic/decoder adaptation (W_k), and output authority ($\beta\boldsymbol{\ell}$) are distinct.

C. Structural Properties and Their Scope

Proposition 1 (Bounded internal states)**.** *With the nonnegative conductance parameters in Table I, $r_{j,0} \in [0,1]$, and $0 < h \le \tau_r$, the gate remains in $[0,1]$. The membrane iterate remains in the convex hull of its initial value, v_{reset}, and the interval $[v_{\mathrm{rest}} + I_b, v_{\mathrm{rest}} + I_b + I_g]$.*

Proof. The unclipped gate step is a convex combination of $r_{j,k}$ and $b_{j,k} \in [0,1]$. Since $I_j \in [I_b, I_b + I_g]$ and $c_j \ge 0$, d_j/λ_j is a convex combination of $v_{\mathrm{rest}} + I_j$ and v_{reset}, and (19) one of $v_{j,k}$ and d_j/λ_j; induction proves both statements. □

Proposition 2 (Bounded authority and baseline recovery)**.** *For $\boldsymbol{u}_{0,k} \in \mathcal{U}$ and any finite W_k, $\boldsymbol{u}_k \in \mathcal{U}$ and $|\delta_{k,i}| \le \beta \ell_i$. If $\beta = 0$, the requested baseline command is recovered at every step. If $W_k = 0$ at a particular step, it is recovered at that step, but learning need not keep $W_{k+1} = 0$.*

Proof. The outer clip enforces membership in $\mathcal{U}$. Its componentwise nonexpansiveness and $\mathrm{sat}_{\ell_i}(u_{0,k,i}) = u_{0,k,i}$ give $|\delta_{k,i}| \le |\beta \ell_i s_{k,i}| \le \beta \ell_i$. The recovery statements follow by substitution. □

With $\beta = 0$ the same allocator and state give the same actuator output. The proposition establishes neither state safety, rotor headroom, stability, nor a bound on tracking improvement; a bounded adversarial correction can still degrade or destabilize a nonlinear plant.

Proposition 3 (Decoder bound and restricted contraction)**.** *If $W_0 \in \mathcal{W}$, then* (22) *keeps $W_k \in \mathcal{W}$. If, additionally, the target and a nonzero activity $\boldsymbol{a}$ are fixed, leak is zero, and readout clipping and projection remain inactive, then*

$$\varepsilon_{k+1} = (1 - \eta \|\boldsymbol{a}\|_2^2)\varepsilon_k. \tag{24}$$

For a nonzero initial error, its norm contracts to zero when $0 < \eta \|\boldsymbol{a}\|_2^2 < 2$.

Proof. Projection gives the first result. Under the additional conditions, $(W_{k+1} - W_k)\boldsymbol{a} = \eta \varepsilon_k \|\boldsymbol{a}\|_2^2$; subtracting this from the fixed target error gives (24). The scalar multiplier has modulus less than one under the stated condition. □

This is the constant-feature PES result [5], not a closed-loop convergence theorem. Since $\|\boldsymbol{a}\|^2 \le 2m$, $0 < \eta < 1/m$ suffices for that restricted model; changing targets, loss of excitation, clipping, and leakage prevent direct application to the experiment, and one fixed regressor does not identify all decoder weights.

D. Allocation-Aware Local Stability

Let $E = [\mathbf{0}_{3\times 1}, I_3]^\mathsf{T}$, $H = E^\mathsf{T}$, and let $\mathcal{A}_T(\boldsymbol{u}) = HB\,\mathrm{clip}_{[0,\boldsymbol{f}_{\max}]}(B^{-1}[T, \boldsymbol{u}^\mathsf{T}]^\mathsf{T})$ be the realized torque at collective T. For two commands at the *same state and collective*, scalar clipping has secant slopes $d_i \in [0,1]$. Consequently,

$$\mathcal{A}_T(\boldsymbol{u}) - \mathcal{A}_T(\boldsymbol{u}_0) = HBDB^{-1}E\boldsymbol{\delta}, \tag{25}$$

$$\|\mathcal{A}_T(\boldsymbol{u}) - \mathcal{A}_T(\boldsymbol{u}_0)\|_2 \le L_A \beta \|\boldsymbol{\ell}\|_2, \tag{26}$$

where $D = \mathrm{diag}(d_i)$ and $L_A = \max_{D\in\mathcal{D}} \|HBDB^{-1}E\|_2$ over the set $\mathcal{D}$ of sixteen diagonal matrices with entries in $\{0,1\}$; by convexity the vertex maximum bounds all intermediate D. The bound holds under rotor clipping, but it neither compares two different closed-loop trajectories nor bounds the height error.

For a constant reference and a smooth shortest-path branch, let $\boldsymbol{x} = [\boldsymbol{e}_q^\mathsf{T}, \boldsymbol{\omega}^\mathsf{T}]^\mathsf{T}$ and $K_q = \mathrm{diag}(\boldsymbol{k}_q)$, $K_\omega = \mathrm{diag}(\boldsymbol{k}_\omega)$. Write the continuous-time attitude subsystem as

$$\dot{\boldsymbol{x}} = A\boldsymbol{x} + \boldsymbol{g}(\boldsymbol{x}) + F\boldsymbol{d},$$
$$A = \begin{bmatrix} 0 & \frac{1}{2}I_3 \\ -\boldsymbol{J}^{-1}K_q & -\boldsymbol{J}^{-1}K_\omega \end{bmatrix}, \qquad F = \begin{bmatrix} 0 \\ \boldsymbol{J}^{-1} \end{bmatrix}, \tag{27}$$

where $\boldsymbol{d} = \boldsymbol{\tau}_{\mathrm{act}} + \boldsymbol{\tau}_{\mathrm{p}} + \boldsymbol{\tau}_{\mathrm{d}} + K_q\boldsymbol{e}_q + K_\omega\boldsymbol{\omega}$ is the total deviation from ideal PD torque, including allocation and command-saturation errors, not just the neural request. The nonlinear remainder is

$$\boldsymbol{g}(\boldsymbol{x}) = \begin{bmatrix} \frac{1}{2}[(\sqrt{1 - \|\boldsymbol{e}_q\|^2} - 1)I_3 + [\boldsymbol{e}_q]_\times]\boldsymbol{\omega} \\ -\boldsymbol{J}^{-1}[\boldsymbol{\omega}]_\times \boldsymbol{J}\boldsymbol{\omega} \end{bmatrix} = O(\|\boldsymbol{x}\|^2), \tag{28}$$

with $[\boldsymbol{z}]_\times \boldsymbol{y} = \boldsymbol{z} \times \boldsymbol{y}$. Norms below use fixed coordinates with angular rates expressed in rad/s.

Theorem 1 (Conditional local practical stability)**.** *Let $\boldsymbol{J}, K_q, K_\omega \succ 0$, let q_d be constant, and use the smooth branch in* (27)*. Choose $Q_L \succ 0$ and the solution $P \succ 0$ of $A^\mathsf{T}P + PA = -Q_L$. On $\|\boldsymbol{x}\| \le r < 1$, suppose $\|\boldsymbol{g}(\boldsymbol{x})\| \le L_r\|\boldsymbol{x}\|^2$ and choose r so that $c = \lambda_{\min}(Q_L) - 2\|P\|L_r r > 0$. Set $p_- = \lambda_{\min}(P)$, $p_+ = \lambda_{\max}(P)$, $b = \|PF\|$, and $V = \boldsymbol{x}^\mathsf{T}P\boldsymbol{x}$. If*

$$V(0) \le p_- r^2, \qquad \|\boldsymbol{d}(t)\| \le \bar{d} < \frac{cr}{2b}\sqrt{\frac{p_-}{p_+}}, \tag{29}$$

then $V \le p_- r^2$ is forward invariant and

$$V(t) \le e^{-\alpha t}V(0) + \frac{4p_+b^2}{c^2}\bar{d}^2, \qquad \alpha = \frac{c}{2p_+}. \tag{30}$$

Thus attitude regulation is locally input-to-state stable; zero perturbation gives local exponential convergence of the plant state.

Proof. The linear system has energy $\frac{1}{2}\boldsymbol{\omega}^\mathsf{T}\boldsymbol{J}\boldsymbol{\omega} + \boldsymbol{e}_q^\mathsf{T}K_q\boldsymbol{e}_q$ with derivative $-\boldsymbol{\omega}^\mathsf{T}K_\omega\boldsymbol{\omega}$. Its only invariant zero-derivative trajectory is the origin, so A is Hurwitz and $P \succ 0$ exists. Equations (27)–(28) give

$$\dot{V} \le -c\|\boldsymbol{x}\|^2 + 2b\|\boldsymbol{x}\|\bar{d} \le -\frac{c}{2}\|\boldsymbol{x}\|^2 + \frac{2b^2}{c}\bar{d}^2.$$

On $V = p_- r^2$, $\|\boldsymbol{x}\| \ge r\sqrt{p_-/p_+}$, so (29) makes the first bound strictly negative. The ellipsoid stays inside the radius-r ball; integration of $\dot{V} \le -\alpha V + 2b^2\bar{d}^2/c$ gives (30). □

If the same-state baseline deviation from ideal PD is bounded by $\bar{d}_0$, (26) permits $\bar{d} = \bar{d}_0 + L_A\beta\|\boldsymbol{\ell}\|$ in the theorem, *provided* (29) holds: a sufficient local robustness condition, independent of the PES trajectory, not a PES convergence proof. A is numerically Hurwitz for both implemented gain sets; the smallness condition is not certified for

the experimental $\beta = 0.5$. Moving references, the 180° sign boundary, sampled-data/solver error (which would need its own bound within $\boldsymbol{d}$), and full translational stability remain outside this result.

a) Nengo realization.: In Nengo 4.1.0 a custom `NeuronType` exposes membrane and gate states, and direct current injection into `ensemble.neurons` bypasses NEF gain/intercept fitting; unfiltered CMG activity uses $\eta_N = \eta(2m)/h$ and Nengo error $\boldsymbol{s}-\boldsymbol{y}^*$, leak uses pre-step weights, and projection follows the simulator step. Its teaching graph adds a one-step latency, so the weight sequences differ; over ten direct/Nengo cases the maximum relative mean-error difference is 2.013×10^{-6}, a tested agreement, not formal equivalence.

IV. Experimental Protocol and Cross-Mission Results

A. Models, Parameters, and Measurements

MuJoCo [14] simulates two Menagerie-derived multirotor models [15], CF2 and X2, as free-flying rigid bodies with site-applied wrenches; the attitude loop commands three body torques and a separate height loop commands collective thrust.

The upstream CF2 model exposes abstract moment channels [16]; we replace them with four rotor-force channels and the shared clipped inverse allocator (11) to expose thrust/torque coupling and per-rotor limits. Rotor ordering is $(-x,-y),(-x,+y),(+x,+y),(+x,-y)$ with reaction signs $(-,+,-,+)$. The CF2 $1.5\times$ scale (Table I) raises a model limit, not measured hardware capability. Both models are simplified (no motor lag, RPM, or battery states, and no validated rotor aerodynamics; a historical CF2 motor step took about 180 ms [17]); feedback uses `qpos`/`qvel`, with noise injected only in Sec. IV-C.

CF2's body axes are principal axes. X2's geometry-derived body tensor, $\boldsymbol{J} = 10^{-2}[3.66517, 0, -.21000;\ 0, 2.54117, 0;\ -.21000, 0, 6.05280]\,\mathrm{kg\,m^2}$ (rows separated by semicolons), differs from both $\mathrm{diag}(\boldsymbol{j}_{\mathrm{tune}})$ and a manufacturer-identified tensor; torque limits use $\boldsymbol{j}_{\mathrm{tune}}$ (Section II).

Each PD/CMG pair shares initial state, reference, duration, allocator, model, and timestep. The primary campaign is nominal and deterministic (no injected disturbances or sensor noise), one run per cell, learning active throughout. We report $\bar{\theta}_q$ and the reduction $100(\bar{\theta}_{q,\mathrm{PD}} - \bar{\theta}_{q,\mathrm{CMG}})/\bar{\theta}_{q,\mathrm{PD}}$ over all 90,000 samples, even when plotted traces are decimated.

The five reference triples (radians) are: Roll $(0.25\sin 0.7t, 0, 0)$; Full3 $0.5\sin(t + (0,\pi/2,\pi)^{\mathsf{T}})$ componentwise; Hard45 $(0.55\sin 0.75t + 0.20\sin(2.10t + 0.40),\ 0.45\sin(0.95t + \pi/3) + 0.18\sin 1.75t,\ 0.65\sin(0.55t+\pi)+0.25\sin(1.35t+0.20))$. Hardstep holds zero for 1 s, then cycles every 2 s through $(.75,-.45,.55)$, $(-.65,.55,-.70)$, $(.50,.50,.90)$, and $(-.80,-.35,.20)$. Yaw360 is $(0, 0, 2\pi(3\xi^2 - 2\xi^3))$ with $\xi = t/45$. Hover is omitted: its near-zero error admits no meaningful percentage.

TABLE I

Model/interface parameters with manufacturer context (top) and canonical 3-channel CMG–PES parameters (bottom). Mass agrees with published CF2/X2 specifications [18], [19]; force limits, inertias, and reaction coefficients are simulation parameters, not identified hardware ratings. Neural states/currents are normalized model variables; learning coefficients are per step.

Parameter	CF2 rotor model	X2 model
Mass M (kg)	0.027	1.325
Rotor $\lvert p_x\rvert, \lvert p_y\rvert$ (m)	$0.0325, 0.0325$	$0.14, 0.18$
Reaction $\lvert\kappa_i\rvert$ (m)	0.005	0.0201
Rotor/total force limit (N)	$.0875 \to .13125$ / $.350 \to .525$	13 / 52
Thrust/weight	1.982	4.001
$\boldsymbol{j}_{\mathrm{tune}}$ ($\mathrm{kg\,m^2}$)	$10^{-5}(2.3951, 2.3951, 3.2347)$	$10^{-2}(6.0711, 3.6468, 2.5412)$
Integrator, timestep	RK4, 0.0005 s	Euler, 0.0005 s
Mfr. thrust data	.568 N[a]	Not published

[a]Historical CF2 total at 93.75% PWM on an external supply [20], not a continuous battery rating; the modeled .525 N is below it.

CMG–PES	Value		Value
h, duration	0.0005 s, 45 s	$\alpha_h, \gamma, g_0, g_1$	$50, 2, 120, 30$
τ, τ_r	0.05 s, 0.005 s	I_b, I_g, G	$0.75, 1.25, 10^5$
$v_{\mathrm{rest}}, v_{\mathrm{reset}}, v_{\mathrm{th}}$	$0, 0, 1$	β, η	$0.5, 8 \times 10^{-4}$
$v_{j,0}, r_{j,0}, \chi$	$0, 0, 1$	$\lambda_W, W_{\max}, k_e, k^*_\omega$	$10^{-6}, 6, 8, 0$

TABLE II

Mean quaternion-angle error [rad] and reduction vs PD [%], 45 s, $n = 1$ per clean cell. PES: canonical Nengo CMG–PES; Fixed: same units, $W_k = W_0$; Mod./Real.: CMG–PES at the two noise levels of Sec. IV-C, two-seed means. CF2 is the $1.5\times$ rotor-authority design.

		Error [rad]		Reduction vs PD [%]			
Model	Mission	PD	CMG–PES	PES	Fixed	Mod.	Real.
X2	Roll	.050565	.041785	17.36	1.93	32.3	32.2
	Full3	.303275	.131748	56.56	9.74	56.4	55.7
	Hard45	.294082	.138007	53.07	10.27	52.9	52.3
	Hardstep	.454528	.359355	20.94	0.46	20.8	20.6
	Yaw360	.075863	.029902	60.58	0.12	56.8	47.8
CF2	Roll	.051337	.042439	17.33	3.31	32.4	32.5
	Full3	.288383	.126476	56.14	4.53	55.9	55.2
	Hard45	.285449	.132319	53.65	6.80	53.4	52.7
	Hardstep	.426182	.347886	18.37	0.42	18.3	18.1
	Yaw360	.075812	.029870	60.60	0.06	56.8	47.8

B. Primary Five-Mission Matrix

CMG–PES reduces mean error in all ten nominal cases under the same actuator interface, most on Yaw360 and Full3 and least on Roll. Hardstep has the largest absolute error and the only nonzero rotor-bound occupancy S (% of samples with a rotor command at a bound, PD/CMG): $.016/.239$ on X2 and $.137/.264$ on CF2. S counts post-clipping commands near a limit, not requested-wrench infeasibility. The results establish deterministic nominal performance relative to this fixed PD law, not optimality over classical controllers; improvement refers to the mean metric and need not hold at every instant.

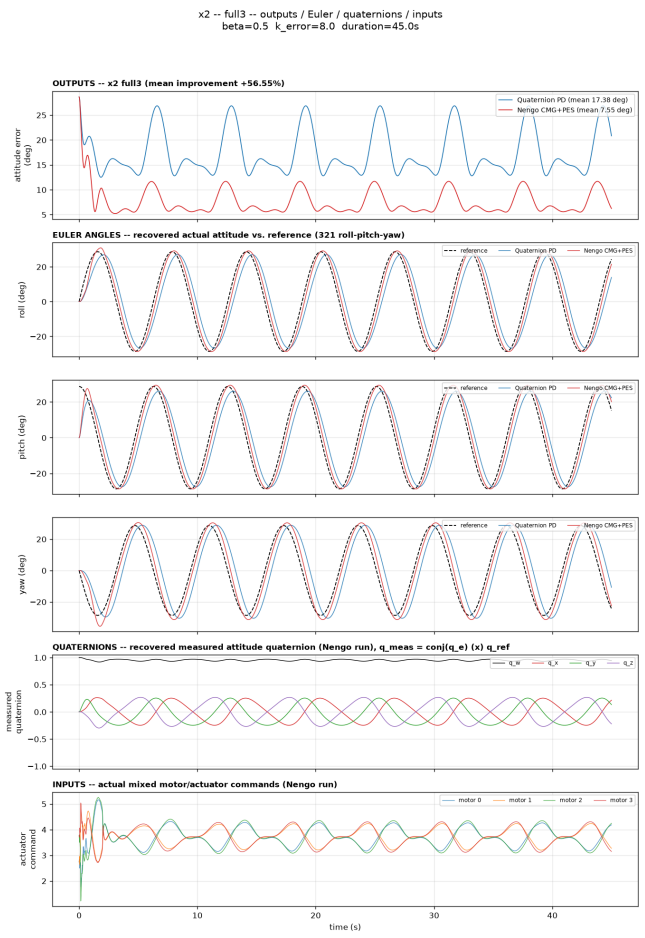


Fig. 4. X2 Full3: PD versus CMG–PES over the 45-s run. Full3 commands all three axes at once and gives the largest reduction that is also noise-insensitive (under one point between the clean and realistic columns of Table II).

C. Sensitivity to Measurement Noise

Each case was rerun with two paired seeds at a moderate ($0.010\,\mathrm{rad}$, $0.025\,\mathrm{rad\,s^{-1}}$) and a realistic ($0.020\,\mathrm{rad}$, $0.050\,\mathrm{rad\,s^{-1}}$) attitude/rate noise level, not calibrated sensor specifications: at every step, zero-mean Gaussian samples form a small Euler-angle quaternion left-multiplied into q and corrupt each rate component. The runner evaluates θ_q from the *corrupted* quaternion used by the controller, not a ground-truth attitude, so the Mod./Real. columns of Table II measure noisy-feedback error. All noisy reductions stay positive, spanning 18.08–55.75% at the higher level; Yaw360 loses about 12.8 pp against clean feedback while Roll's increases. Seed SDs are .001–.101 pp (moderate) and .002–.031 pp (realistic): limited repeatability, not a population confidence interval.

D. Ablation: What Drives the Improvement?

We separate decoder adaptation (PES), input-dependent conductance modulation (χ), and readout gain (G). Toggling χ, which retains the base shunt $g_0 r_j^2$, is numerically indistinguishable with the decoder mode held fixed. *Fixed versus adaptive decoder.* Holding $W_k = W_0$ keeps the neuron dynamics but reduces the readout to $\boldsymbol{a}^+ - \boldsymbol{a}^-$, a bounded, stateful image of the PD request. It still helps (X2 0.12–10.27%, CF2 0.06–6.80%; Table II), so fixing the decoder does not remove *all* improvement, yet it recovers at most a fifth of the PES gain in any cell. Fig. 5 shows why, row by row. *Membrane and gate.* At $I_j = I_b$ the membrane rests at $v_j \approx 0.75 < v_{\mathrm{th}}$, so $r_j = \sigma(\alpha_h(I_b - v_{\mathrm{th}})) = \sigma(-12.5) \approx 3.7 \times 10^{-6}$: the gate is deep in its logistic tail, not closed. Once $r_j \geq 1/2$, $\gamma = 2$ and $v_j \geq 0$ (Proposition 1) give $b_{j,k} \geq 1/2$ in (18), so an opened gate cannot close (none did in the twenty runs), and the shunt holds v_j near 0.09. *Activity.* With $G = 10^5$ the resting gate already gives $a_j \approx 0.37$, graded by the input; a latched gate pins $a_j = 1$. With the fixed decoder only the $\pm y$ pair latches on Full3 (both models), so $a_y^+ - a_y^- = 0$ and the pitch residual is exactly zero after 1 s while the sub-threshold units carry roll and yaw. Under PES on X2, $\boldsymbol{a}_k \equiv \mathbf{1}$ from $2.02\,\mathrm{s}$ (all gates latched by $2.4\,\mathrm{s}$), and for constant $\bar{\boldsymbol{a}}$ with inactive projection the direct update gives $\boldsymbol{z}_k = W_k \bar{\boldsymbol{a}}$ the recursion

$$\boldsymbol{z}_{k+1} = (1 - \lambda_W)\boldsymbol{z}_k + \eta \|\bar{\boldsymbol{a}}\|^2 [\boldsymbol{y}_k^* - \mathrm{clip}_{[-1,1]}(\boldsymbol{z}_k)], \quad (31)$$

the leaky constant-feature regime of Proposition 3: a first-order filter of the teaching target with pole 0.9952 (about $0.10\,\mathrm{s}$) while the readout clip is inactive. *Residual and decoder.* The residual RMS rises from 0.026 to $0.123\,\mathrm{N\,m}$ and the decoder moves $\|W_T - W_0\|_F = 0.37$; on Roll, Full3, Hard45, and Hardstep PES applies 4–$12\times$ the fixed residual RMS on both models. Where no gate opens (Roll, Yaw360; $r_j \leq 1.6 \times 10^{-3}$) PES lifts the driven membranes toward threshold ($v_j \leq 0.87$) and uses the graded tail. Decoder adaptation is therefore the principal lever: the CMG dynamics supply a bounded basis, latched or graded, on which PES shapes the correction.

E. Neuron-Family Comparison

The Nengo benchmark covers two models, five missions, and eleven controllers: PD and fixed/PES variants of CMG, LIF, ALIF, Izhikevich [21], and LIFRate, all sharing the bounded residual, saturation, and allocation interface; only the neuron dynamics and native-state-to-activity map differ. Fig. 6 shows two patterns. First, PES is the dominant lever in every family: fixed decoders rarely match their PES counterparts, even with more expressive dynamics, extending Sec. IV-D to every basis. Second, lower error is not free: some Izhikevich–PES cells reach the smallest error, but at an average normalized residual RMS near 0.37 versus 0.12 for CMG–PES and with larger decoder movement, so equal output caps do not imply equal realized effort. Sparse LIF/ALIF activity in some cells limits the effective basis here, which can blunt learning without implying those families cannot learn under other encodings or gains. CMG–PES thus attains comparable tracking at markedly lower realized effort and more conservative decoder motion.

F. Disturbance Rejection and Retention

An equilibrium test applies $\boldsymbol{\tau}_{\mathrm{d}} = d_b \rho(t)[\sin(1.31t + .2), .8\sin(1.73t + 1.1), .6\sin(2.17t + 2)]^{\mathsf{T}}$ in body axes. After

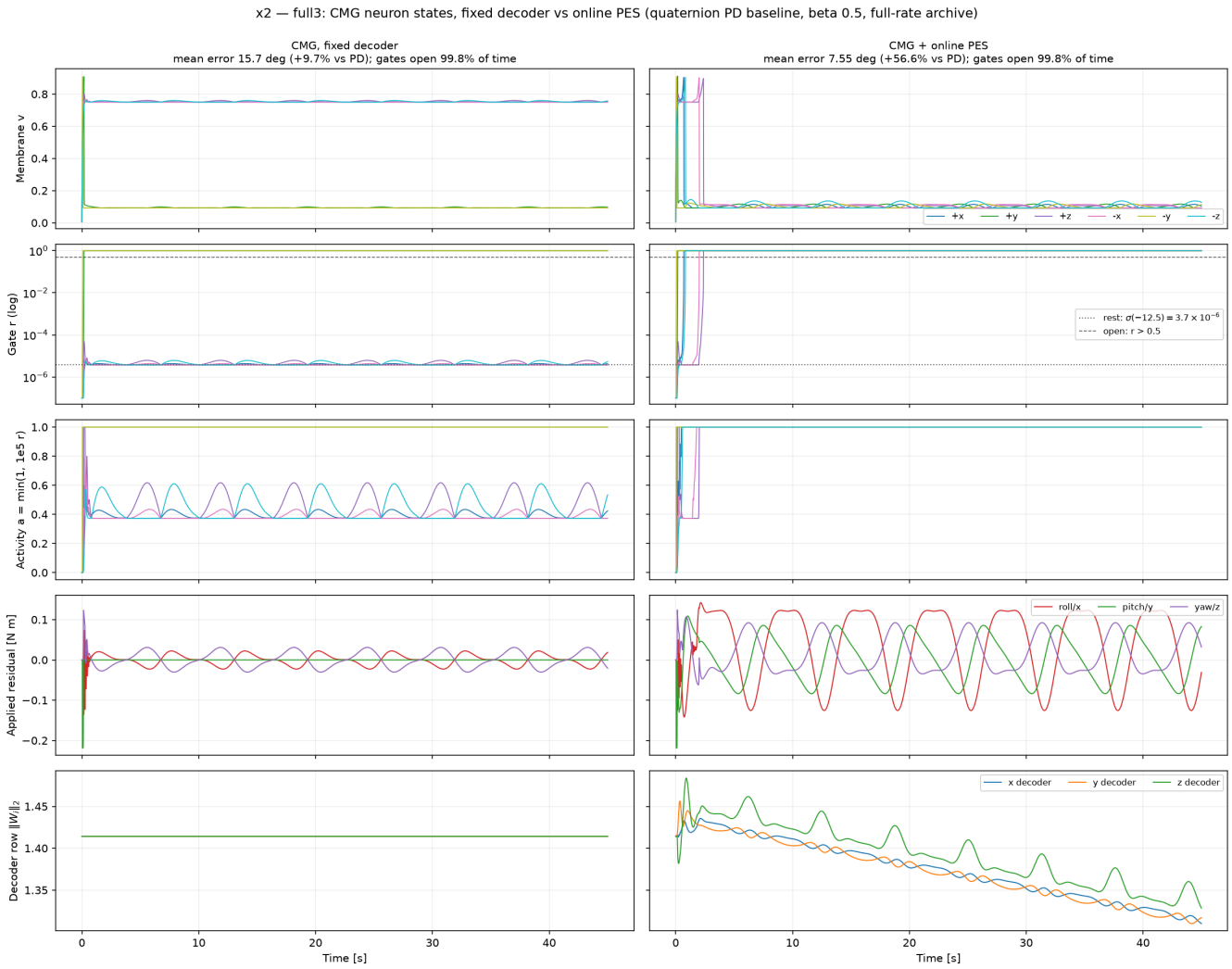


Fig. 5. X2 Full3 internal CMG signals on the trajectory of Fig. 4: fixed decoder $W_k = W_0$ (left) versus online PES (right), 2-kHz archive. Rows: membrane v_j; gate r_j (log; dotted: rest $\sigma(\alpha_h(I_b - v_{\text{th}}))$, dashed: $1/2$); activity a_j; requested residual $\boldsymbol{\delta}$ (pre-allocation, not rotor clipping); decoder norms $\|W_i\|_2$.

5 s of quiet, ρ alternates 5-s on/off intervals with 0.25-s raised-cosine edge ramps. For X2, $d_b = .15$ N m is the first component's amplitude, not a vector-norm bound. The four-burst 45-s experiment (Fig. 7) cuts burst-phase mean error from $11.50°$ to $6.65°$ (42.2%) and post-burst quiet error from $1.21°$ to $.99°$ (18.0%). The CF2 rerun uses $d_b = .00015$ N m (a selected scaling, not an exact inertia-ratio match); its full-run mean falls from .141199 to .070259 rad (50.24%).

V. Discussion and Conclusion

A bounded residual of six continuous CMG units and eighteen PES-adapted decoder weights augments a fixed quaternion PD baseline without changing its gains or actuator interface, reducing mean quaternion-angle error by 17.3–60.6% over ten matched 45-s MuJoCo cases on two modeled platforms, within rotor saturation and with exact baseline recovery at zero authority; direct and Nengo realizations agree to 2.013×10^{-6} relative mean error.

Decoder adaptation is the dominant source of improvement in the tested regime: a fixed decoder gives at most 10.3%, and toggling χ is indistinguishable. Opened gates latch and saturate the readout, where a fixed paired decoder can cancel an axis, while sub-threshold units stay in a graded logistic tail; PES shapes the correction on either basis. Across all five neuron families PES is the consistent differentiator, and the dynamics mainly set an effort–accuracy tradeoff. The readout gain G, which sets the resting activity at 0.37, is a key sensitivity parameter.

Limitations: results are in simulation (Sec. IV-C bounds noise sensitivity; model mismatch and hardware effects are not addressed), the retention horizon is limited, and bounded commands, states, and weights are implementation properties, not global stability proofs.

Future work includes baselines with desired-rate feedforward and integral action, matched-effort comparisons against non-neural bounded bases, sweeps of G and χ, longer retention studies, held-out evaluations with frozen learned



Fig. 6. Neuron-family benchmark, X2 (top) and CF2 (bottom): mean q-error improvement over PD (left) and normalized residual effort, RMS residual over the per-axis limit (right). Rows are family and learning mode, columns missions; all cells share the bounded residual, saturation, and allocation interface. On both platforms PES, not the neuron model, separates improving rows from flat ones, and families matching or exceeding CMG–PES on error spend three to twenty times its effort.

weights, full 6-DOF extension, and, most importantly, validation on physical aerial, ground, and other robotic platforms; the same bounded-residual wrapper extends to nonlinear and optimal-control baselines with hard actuator constraints.

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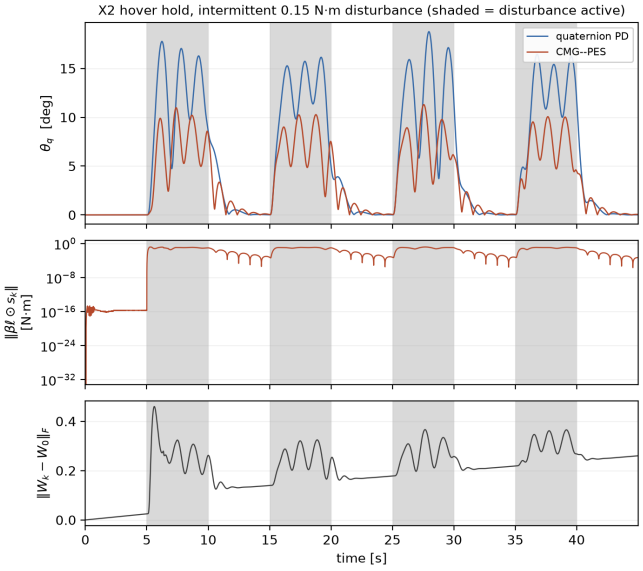


Fig. 7. X2 hover hold under four $0.15\,\mathrm{N\,m}$ bursts (shaded): attitude error (top), residual magnitude, log scale (centre), and decoder displacement from W_0 (bottom). Before the first burst the residual is at numerical precision: negligible correction, not zero computation or energy cost.

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